

# OS

## ✔ Unit 1 MCQ Test (Computer Fundamentals & Operating Systems)

### Section A: Generations of Programming Languages + Compiler

**Q1. Which generation of programming language is written purely in binary (0s and 1s)?**

- A) 2G
- B) 3G
- C) 1G
- D) 4G

✔ **Answer: C) 1G**

**Explanation:** 1G = machine language, binary instructions, hardware dependent.

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**Q2. Assembly language belongs to which generation?**

- A) 1G
- B) 2G
- C) 3G
- D) 5G

✔ **Answer: B) 2G**

**Explanation:** 2G uses mnemonics like MOV, ADD and is still hardware dependent.

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**Q3. The biggest issue in 1G and 2G languages was:**

- A) Very high runtime memory usage
- B) No OS support
- C) Hardware dependency and lack of portability
- D) Slow I/O operations

✔ **Answer: C) Hardware dependency and lack of portability**

**Explanation:** Programs worked only on specific hardware architectures.

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**Q4. A compiler is a translator that converts:**

- A) Binary to assembly
- B) High-level language to machine language
- C) Machine language to high-level language
- D) Assembly to SQL

✔ **Answer: B) High-level language to machine language**

**Explanation:** Compiler converts HLL source code into machine-level code.

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**Q5. Which of the following is an example of a 4G language?**

- A) SQL
- B) Assembly
- C) Machine code
- D) C

☒ **Answer: A) SQL**

**Explanation:** 4G languages are task-oriented and domain-specific (DB, analytics).

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**Q6. 5G programming languages are mostly related to:**

- A) Networking protocols
- B) Hardware drivers
- C) Artificial Intelligence and Natural Language Processing
- D) Database normalization

☒ **Answer: C) Artificial Intelligence and Natural Language Processing**

**Explanation:** 5G focuses on intelligent/self-learning systems.

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## **Section B: Operating System Basics + OS vs Application**

**Q7. An Operating System is mainly considered as:**

- A) A hardware device
- B) System software that manages hardware resources
- C) A compiler
- D) A programming language

☒ **Answer: B) System software that manages hardware resources**

**Explanation:** OS acts as a bridge between user and hardware.

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**Q8. Which of the following is a correct conceptual flow?**

- A) User → Kernel → OS → Driver → Device
- B) User → OS → Kernel → Driver → Device
- C) User → Device → Driver → OS → Kernel
- D) User → Compiler → OS → Kernel → Device

☒ **Answer: B) User → OS → Kernel → Driver → Device**

**Explanation:** User interacts with OS, kernel handles system calls, drivers talk to device.

---

**Q9. The core part of OS that directly manages hardware is:**

- A) API
- B) Shell

- C) Kernel
- D) GUI

✓ **Answer: C) Kernel**

**Explanation:** Kernel is the central component that controls hardware resources.

---

**Q10. Which is mandatory for the system operation?**

- A) Excel
- B) Chrome
- C) VLC Player
- D) Operating System

✓ **Answer: D) Operating System**

**Explanation:** OS is essential. Applications are optional.

---

**Q11. Which one is an example of Application Software?**

- A) Linux
- B) Android
- C) Windows
- D) Google Chrome

✓ **Answer: D) Google Chrome**

**Explanation:** Chrome runs on top of an OS → application software.

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### **Section C: OS Components + Basic Computer Organization**

**Q12. Which OS component is responsible for allocating and managing RAM?**

- A) Device management
- B) File system
- C) Memory management
- D) Networking

✓ **Answer: C) Memory management**

**Explanation:** It controls allocation, deallocation, and efficient RAM usage.

---

**Q13. Which OS component manages drivers and I/O devices?**

- A) Process Management
- B) Device & I/O Management
- C) Security Management
- D) CPU Scheduling

✓ **Answer: B) Device & I/O Management**

**Explanation:** Handles device controllers, drivers, input/output operations.

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**Q14. Which of the following is NOT a basic computer component OS works with?**

- A) CPU
- B) RAM
- C) System Bus
- D) Compiler

☒ **Answer: D) Compiler**

**Explanation:** Compiler is a translator program, not a hardware component.

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#### **Section D: Types of Operating Systems + RTOS**

**Q15. Which OS is designed mainly for PCs and laptops?**

- A) Embedded OS
- B) Desktop OS
- C) RTOS
- D) Server OS

☒ **Answer: B) Desktop OS**

**Explanation:** Desktop OS is for personal computing (Windows, Ubuntu Desktop).

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**Q16. Which OS is optimized for multi-user and server performance?**

- A) Server OS
- B) Mobile OS
- C) Desktop OS
- D) Batch OS

☒ **Answer: A) Server OS**

**Explanation:** Server OS supports high loads, multiple users, services.

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**Q17. RTOS is mostly used where response time must be predictable, such as:**

- A) Playing games
- B) Video editing
- C) Aircraft systems / medical devices
- D) Watching movies

☒ **Answer: C) Aircraft systems / medical devices**

**Explanation:** RTOS is used in critical systems where deadlines matter.

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**Q18. In Hard RTOS:**

- A) Deadlines may be missed occasionally
- B) Deadlines must never be missed

- C) Only GUI programs run
- D) It only supports multitasking

☒ **Answer: B) Deadlines must never be missed**

**Explanation:** Hard RTOS = strict deadline guarantee.

---

**Q19. In Soft RTOS:**

- A) Missing a deadline causes system failure always
- B) Deadlines are strict like Hard RTOS
- C) Occasional delays are acceptable
- D) It cannot handle interrupts

☒ **Answer: C) Occasional delays are acceptable**

**Explanation:** Soft RTOS tolerates delays but performance reduces.

---

**Section E: User Mode vs Kernel Mode**

**Q20. Application programs generally run in:**

- A) Kernel Mode
- B) User Mode
- C) BIOS Mode
- D) Debug Mode

☒ **Answer: B) User Mode**

**Explanation:** User mode has limited access to protect the system.

---

**Q21. Which mode has full access to hardware resources?**

- A) User Mode
- B) Kernel Mode
- C) Guest Mode
- D) Batch Mode

☒ **Answer: B) Kernel Mode**

**Explanation:** Kernel mode allows privileged instructions and device control.

---

**Q22. Switching from user mode to kernel mode occurs using:**

- A) Compiler
- B) System calls or interrupts
- C) Cache memory
- D) BIOS only

☒ **Answer: B) System calls or interrupts**

**Explanation:** System call triggers kernel service, interrupts also switch context.

---

## Section F: Interrupts + System Calls

**Q23. An interrupt is best defined as:**

- A) A storage device
- B) A signal that pauses current execution to handle an event
- C) A database command
- D) A high-level instruction only

☒ **Answer: B) A signal that pauses execution**

**Explanation:** CPU stops current work and handles event like I/O completion.

---

**Q24. System calls are used for:**

- A) Converting HLL to machine code
- B) Requesting OS services from kernel
- C) Designing UI
- D) Formatting hard disks only

☒ **Answer: B) Requesting OS services from kernel**

**Explanation:** Example: open(), read(), write(), fork().

---

**Q25. Which of the following is a system call example?**

- A) sum()
- B) print()
- C) fork()
- D) addTwoNumbers()

☒ **Answer: C) fork()**

**Explanation:** fork() is a system call used to create a new process.

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## Section G: OS Services + User Interfaces

**Q26. Which OS service provides CLI/GUI interaction?**

- A) File management
- B) User interface
- C) Memory management
- D) Device management

☒ **Answer: B) User interface**

**Explanation:** OS provides CLI, GUI, Batch, AUI interfaces.

---

**Q27. GUI stands for:**

- A) General User Input
- B) Graphical User Interface
- C) Global User Interaction
- D) General Utility Interface

✅ **Answer: B) Graphical User Interface**

**Explanation:** GUI uses visual components (windows, icons, mouse).

---

**Q28. In a CLI interface:**

- A) Commands are typed manually
- B) Commands are stored in file
- C) Mouse-based interaction happens
- D) Only voice input is used

✅ **Answer: A) Commands are typed manually**

**Explanation:** CLI = command-line interface.

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**Q29. Batch interface means:**

- A) Voice-based execution
- B) Commands stored in a file and executed as a batch
- C) Mouse-based interface
- D) Only works in kernel mode

✅ **Answer: B) Commands stored and executed**

**Explanation:** Batch systems run a set of predefined commands.

---

**Q30. AUI is best described as:**

- A) Audio User Interface (voice-based)
- B) Advanced User Interrupt
- C) Automatic User Input
- D) Application UI

✅ **Answer: A) Audio User Interface**

**Explanation:** AUI involves voice-based interaction.

---

## **Section H: Communication Types**

**Q31. IPC stands for:**

- A) Internal Process Control
- B) Inter-Process Communication
- C) Input Process Communication
- D) Integrated Program Control

☒ **Answer: B) Inter-Process Communication**

**Explanation:** IPC allows different processes to exchange data.

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**Q32. Which communication happens within the same process?**

- A) Broadcast
- B) Inter-process communication
- C) Intra-process communication
- D) Message-oriented only

☒ **Answer: C) Intra-process communication**

**Explanation:** Intra-process is internal within one process.

---

### **Section I: Process Management + Process States**

**Q33. Correct process state order is:**

- A) New → Running → Runnable → Blocked → Terminated
- B) New → Runnable → Running → Blocked → Terminated
- C) Running → New → Blocked → Terminated
- D) Runnable → New → Running → Terminated

☒ **Answer: B) New → Runnable → Running → Blocked → Terminated**

**Explanation:** Standard state flow.

---

**Q34. A process waiting for I/O completion is in:**

- A) Running
- B) Runnable
- C) Blocked
- D) Terminated

☒ **Answer: C) Blocked**

**Explanation:** Blocked = waiting for resource/event.

---

**Q35. Which states occur only once in a process lifecycle?**

- A) Runnable and Running
- B) Running and Blocked
- C) New and Terminated
- D) Blocked and Runnable

☒ **Answer: C) New and Terminated**

**Explanation:** Creation and exit happen once; others repeat.

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### **Section J: Schedulers + Scheduling Algorithms**



**Q36. Long-term scheduler is responsible for:**

- A) Selecting process from ready queue to CPU
- B) Admitting jobs into ready queue
- C) Managing swapping in/out
- D) Handling device interrupts

☒ **Answer: B) Admitting jobs into ready queue**

**Explanation:** It decides which jobs enter the system.

---

**Q37. Short-term scheduler is responsible for:**

- A) Selecting process for CPU execution
- B) Swapping processes to disk
- C) Loading kernel during booting
- D) Controlling security policies

☒ **Answer: A) Selecting process for CPU**

**Explanation:** Also called CPU scheduler.

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**Q38. Medium-term scheduler is responsible for:**

- A) Loading OS into memory
- B) Swapping and controlling degree of multiprogramming
- C) Assigning files to storage
- D) Running system calls

☒ **Answer: B) Swapping and multiprogramming degree**

**Explanation:** It manages suspended/swapped processes.

---

**Q39. Which is a CPU scheduling algorithm?**

- A) Paging
- B) Segmentation
- C) Round Robin
- D) Archival

☒ **Answer: C) Round Robin**

**Explanation:** RR is a time-slicing CPU scheduling algorithm.

---

**Q40. FCFS stands for:**

- A) First Come First Serve
- B) Fast CPU For Scheduling
- C) First Cache First Storage
- D) File Control For System

☒ **Answer: A) First Come First Serve**

**Explanation:** FCFS schedules in arrival order.

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### **Section K: Memory Management Fundamentals**

**Q41. Correct memory hierarchy order from fastest to slowest:**

- A) HDD → RAM → Cache → Register
- B) Register → Cache → RAM → HDD/SSD
- C) RAM → Cache → Register → HDD
- D) Cache → Register → SSD → RAM

☒ **Answer: B) Register → Cache → RAM → HDD/SSD**

**Explanation:** Speed decreases downward.

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**Q42. As we move upward in memory hierarchy:**

- A) Speed decreases
- B) Cost per bit decreases
- C) Space decreases but speed increases
- D) Space increases but cost increases

☒ **Answer: C) Space decreases but speed increases**

**Explanation:** Registers fastest but smallest; HDD largest but slow.

---

**Q43. Swapping refers to:**

- A) Converting binary to assembly
- B) Moving data between RAM and disk
- C) Cleaning cache memory
- D) Compressing files

☒ **Answer: B) Moving data between RAM and disk**

**Explanation:** Used to free RAM or bring required process pages.

---

**Q44. Cache memory is best described as:**

- A) Permanent storage device
- B) A high-speed buffer memory
- C) A scheduling algorithm
- D) A system call

☒ **Answer: B) A high-speed buffer memory**

**Explanation:** Cache stores frequently used data for fast access.

---

**Q45. Temporal locality means:**

- A) Programs access nearby memory locations
- B) Same data is reused within short time
- C) Data is stored only in disk
- D) CPU accesses random memory always

✅ **Answer: B) Same data reused soon**

**Explanation:** If accessed once, likely accessed again shortly.

---

**Q46. Spatial locality means:**

- A) Same data reused soon
- B) Nearby memory locations are likely accessed
- C) CPU always uses registers only
- D) Disk is faster than RAM

✅ **Answer: B) Nearby memory locations accessed**

**Explanation:** If one address used, nearby addresses may be used.

---

**Q47. Paging uses:**

- A) Variable-size blocks
- B) Fixed-size blocks
- C) Only logical division
- D) No memory division

✅ **Answer: B) Fixed-size blocks**

**Explanation:** Paging divides memory into fixed-size pages/frames.

---

**Q48. Segmentation uses:**

- A) Fixed-size blocks
- B) Logical variable-size divisions
- C) Only physical address mapping
- D) Cache buffers

✅ **Answer: B) Logical variable-size divisions**

**Explanation:** Segments based on program structure (code/data/stack).

---

**Q49. Virtual memory means:**

- A) RAM is ignored
- B) Disk is used as an extension of RAM
- C) CPU becomes virtual
- D) Cache replaces SSD

☒ **Answer: B) Disk acts as RAM extension**

**Explanation:** Gives illusion of large memory using secondary storage.

---

### **Section L: Booting Process**

**Q50. The first step after power on is:**

- A) Kernel loads drivers
- B) BIOS/UEFI performs POST
- C) Init system starts daemons
- D) GUI appears

☒ **Answer: B) BIOS/UEFI performs POST**

**Explanation:** POST checks hardware before OS loads.

---

**Q51. Who loads the kernel into memory?**

- A) Device controller
- B) Bootloader
- C) Compiler
- D) CPU cache

☒ **Answer: B) Bootloader**

**Explanation:** Bootloader loads OS kernel into memory.

---

**Q52. After kernel initialization, which system starts services/daemons?**

- A) Shell
- B) Init system
- C) BIOS
- D) Cache controller

☒ **Answer: B) Init system**

**Explanation:** init/systemd launches background services.

---

**Q53. The final visible boot stage is generally:**

- A) Machine code execution
- B) Login prompt or GUI screen
- C) POST
- D) CPU scheduling starts

☒ **Answer: B) Login prompt or GUI**

**Explanation:** End-user sees login/desktop.

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## **Linux & Shell Scripting — IMP MCQs (Exam Focused)**

### **Linux Basics + Components**

**Q1. Linux is best described as:**

- A) A programming language
- B) A Unix-like open-source OS kernel originally created by Linus Torvalds (1991)
- C) A database system
- D) A Windows utility

 **Answer: B**

**Explanation:** Linux is open-source Unix-like OS, created in 1991.

---

**Q2. Linux system is essentially:**

- A) Kernel + BIOS
- B) Kernel + GNU utilities
- C) Shell + GUI only
- D) Drivers + Hardware

 **Answer: B**

**Explanation:** Kernel manages hardware, GNU provides commands/tools.

---

**Q3. Which component loads the Linux kernel during startup?**

- A) Daemon
- B) Bootloader (GRUB)
- C) Shell
- D) Desktop environment

 **Answer: B**

**Explanation:** Bootloader loads kernel into memory.

---

**Q4. The init system that starts services/daemons in modern Linux is:**

- A) GNOME
- B) GRUB
- C) systemd
- D) bash

 **Answer: C**

**Explanation:** systemd manages services and background daemons.

---

**Q5. Which is the correct execution flow?**

- A) User → Kernel → Shell → Hardware
- B) User → Shell → Kernel → Drivers → Hardware

C) Kernel → User → Hardware → Drivers

D) Shell → User → Kernel → Hardware

✅ **Answer: B**

**Explanation:** Shell interprets commands → kernel executes via drivers.

---

### Kernel Types

**Q6. Which kernel type runs almost all services in one large program and is very fast?**

A) Microkernel

B) Hybrid kernel

C) Monolithic kernel

D) Distributed kernel

✅ **Answer: C**

**Explanation:** Monolithic = high performance but harder to manage.

---

**Q7. Microkernel architecture is best known for:**

A) Being fastest always

B) High modularity (services outside kernel)

C) No security

D) No drivers required

✅ **Answer: B**

**Explanation:** Microkernel keeps minimal functions in kernel.

---

### Linux File System Structure

**Q8. Which directory contains configuration files in Linux?**

A) /tmp

B) /etc

C) /home

D) /var

✅ **Answer: B**

**Explanation:** /etc = system configuration.

---

**Q9. Which directory stores logs and variable data?**

A) /var

B) /usr

C) /tmp

D) /

✓ **Answer: A**

**Explanation:** /var contains logs, spool data, variable files.

---

### **Core Commands (Most Tested)**

**Q10. What does pwd do?**

- A) Prints process details
- B) Shows current directory full path
- C) Displays disk free space
- D) Creates directory

✓ **Answer: B**

**Explanation:** pwd = Print Working Directory.

---

**Q11. . and .. represent:**

- A) Root and home
- B) Current and parent directory
- C) File and folder
- D) Read and write

✓ **Answer: B**

**Explanation:** . current, .. parent directory.

---

**Q12. Which command lists files in long format with permissions?**

- A) ls -a
- B) ls -l
- C) ls -p
- D) ls -x

✓ **Answer: B**

**Explanation:** ls -l shows detailed listing.

---

**Q13. Which command creates an empty file (or updates timestamp)?**

- A) mkdir
- B) cat
- C) touch
- D) mv

✓ **Answer: C**

**Explanation:** touch creates empty file / changes timestamp.

---

**Q14. Which command is mainly used to rename a file?**

- A) cp
- B) mv
- C) rm
- D) ls

✅ **Answer: B**

**Explanation:** mv old new = rename operation.

---

**Q15. What does cat >> file.txt do?**

- A) Deletes file content
- B) Appends content to file
- C) Reads file as read-only
- D) Renames file

✅ **Answer: B**

**Explanation:** >> appends, doesn't overwrite.

---

**Q16. Which command allows backward navigation while viewing content?**

- A) more
- B) less
- C) cat
- D) head

✅ **Answer: B**

**Explanation:** less supports scrolling back/forward.

---

**Q17. To see live updates at the end of a file (log monitoring), use:**

- A) tail -f
- B) head -f
- C) cat -f
- D) wc -f

✅ **Answer: A**

**Explanation:** tail -f follows appended content.

---

### Searching + Filtering (Very Important)

**Q18. Which command searches for a pattern inside file contents?**

- A) find
- B) grep
- C) sort
- D) cut



✓ **Answer: B**

**Explanation:** grep searches text by pattern.

---

**Q19. Which command searches files by name/type inside directories?**

- A) grep
- B) awk
- C) find
- D) tee

✓ **Answer: C**

**Explanation:** find locates files/directories.

---

**Q20. Which command extracts fields using a delimiter (like CSV)?**

- A) sort
- B) cut
- C) sed
- D) wc

✓ **Answer: B**

**Explanation:** cut -d',' -f1 file.csv.

---

**Q21. Which symbol is used for piping output of one command into another?**

- A) &
- B) >
- C) |
- D) \$

✓ **Answer: C**

**Explanation:** command1 | command2.

---

**Q22. tee is used to:**

- A) Delete file and show output
- B) Show output and save it into file simultaneously
- C) Compress files
- D) Kill a process

✓ **Answer: B**

**Explanation:** tee writes output to terminal + file.

---

**Disk/System Info**

**Q23. Which command shows disk free space usage?**

- A) du
- B) df
- C) ps
- D) cal

☒ **Answer: B**

**Explanation:** df -h shows filesystem usage.

---

**Q24. Which command shows directory/file size usage?**

- A) df
- B) du
- C) ls
- D) id

☒ **Answer: B**

**Explanation:** du -h shows disk usage.

---

### Archiving + Networking

**Q25. Which command creates/extracts .tar.gz archives?**

- A) zip
- B) tar
- C) gzip only
- D) mv

☒ **Answer: B**

**Explanation:** tar -cvzf create, tar -xvzf extract.

---

**Q26. Which command checks connectivity to a host?**

- A) netstat
- B) ssh
- C) ping
- D) scp

☒ **Answer: C**

**Explanation:** ping tests network reachability.

---

**Q27. Which command securely logs into remote host?**

- A) scp
- B) ssh
- C) route
- D) nc

✅ **Answer: B**

**Explanation:** ssh user@host for secure remote login.

---

**Q28. Which command securely copies files to a remote machine?**

- A) ssh
- B) scp
- C) netstat
- D) ifconfig

✅ **Answer: B**

**Explanation:** scp = secure copy over SSH.

---

### Process Management

**Q29. Which command shows active processes?**

- A) ps
- B) kill
- C) top
- D) jobs

✅ **Answer: A**

**Explanation:** ps aux lists running processes.

---

**Q30. kill -9 PID means:**

- A) Pause process temporarily
- B) Force kill a process
- C) Restart system
- D) Show job list

✅ **Answer: B**

**Explanation:** -9 sends SIGKILL (force terminate).

---

**Q31. top / htop are used for:**

- A) File search
- B) Real-time system process monitoring
- C) File compression
- D) User creation

✅ **Answer: B**

**Explanation:** Displays CPU, memory usage, and processes.

---

### Users + Permissions (Must Know)

**Q32. Which command changes file permissions?**

- A) chown
- B) chmod
- C) usermod
- D) passwd

☒ **Answer: B**

**Explanation:** chmod modifies permission bits.

---

**Q33. In permission digits, value 7 means:**

- A) Read only
- B) Write only
- C) Execute only
- D) Read + Write + Execute

☒ **Answer: D**

**Explanation:**  $7 = 4 + 2 + 1 = rwx$ .

---

**Q34. chmod 755 file.sh gives permissions:**

- A) Owner=rwx, Group=rx, Others=rx
- B) Owner=rw, Group=r, Others=r
- C) Owner=rwx, Group=rwx, Others=rwx
- D) Owner=rx, Group=rwx, Others=rwx

☒ **Answer: A**

**Explanation:** 7=rwx, 5=rx, 5=rx.

---

**Q35. Which command changes ownership of a file?**

- A) chmod
- B) chown
- C) useradd
- D) groups

☒ **Answer: B**

**Explanation:** chown user:group file.

---

## Shell Scripting Basics

**Q36. The purpose of #!/bin/bash in a script is:**

- A) To comment the program
- B) To specify which shell should execute the script

- C) To give execute permission
- D) To compile shell code

✓ **Answer: B**

**Explanation:** Shebang defines interpreter.

---

**Q37. To run a script using current directory, use:**

- A) file.sh
- B) bash file.sh only
- C) ./file.sh
- D) run file.sh

✓ **Answer: C**

**Explanation:** ./ executes file from current directory.

---

**Q38. Correct syntax to create a variable in shell:**

- A) name = value
- B) name:value
- C) name=value
- D) \$name=value

✓ **Answer: C**

**Explanation:** No spaces allowed around =.

---

**Q39. Which command takes input from user into a variable?**

- A) echo
- B) read
- C) cat
- D) touch

✓ **Answer: B**

**Explanation:** read name stores input into variable.

---

### Operators + Conditions

**Q40. Which numeric test checks “greater than”?**

- A) -eq
- B) -gt
- C) -lt
- D) -ne

✓ **Answer: B**

**Explanation:** -gt is greater-than.

---

**Q41. In shell, exit status 0 means:**

- A) Failure
- B) Success
- C) Warning
- D) System crash

☒ **Answer: B**

**Explanation:** By convention, 0 = success.

---

**Q42. Which variable stores the exit status of the last executed command?**

- A) \$\$
- B) \$?
- C) \$0
- D) \$@

☒ **Answer: B**

**Explanation:** \$? contains last command's exit code.

---

**Positional Parameters (Super IMP)**

**Q43. \$0 represents:**

- A) Number of arguments
- B) All arguments
- C) Script name
- D) First argument

☒ **Answer: C**

**Explanation:** \$0 is the script filename.

---

**Q44. \$# represents:**

- A) Script name
- B) Number of arguments passed
- C) All arguments in single string
- D) Exit status

☒ **Answer: B**

**Explanation:** \$# counts total arguments.

---

**Q45. Difference between \$\* and \$@ is:**

- A) Both are identical always
- B) \$\* treats all args as a single string, \$@ treats them separately

- C) \$@ is exit status
- D) \$\* is argument count

✓ **Answer: B**

**Explanation:** Very common MCQ/trap.

---

### Loops + Case

#### Q46. while loop runs:

- A) Until condition becomes true
- B) While condition is true
- C) Only once
- D) Only in GUI

✓ **Answer: B**

**Explanation:** while continues while condition is true.

---

#### Q47. until loop runs:

- A) Until condition becomes true
- B) While condition is true
- C) Forever always
- D) Only for arrays

✓ **Answer: A**

**Explanation:** until continues until condition becomes true.

---

#### Q48. Case statement in shell is mainly used for:

- A) Looping
- B) Multiple condition selection (switch-like)
- C) File deletion
- D) Memory allocation

✓ **Answer: B**

**Explanation:** case variable in ... esac.

---

### ACL + System Variables

#### Q49. ACL is used when:

- A) chmod provides enough control always
- B) Normal file permissions are insufficient
- C) Only root can login
- D) No groups exist

✅ **Answer: B**

**Explanation:** ACL allows user-specific/group-specific permissions.

---

**Q50. PATH environment variable contains:**

- A) The current directory only
- B) Directories searched for commands
- C) Password file location
- D) CPU usage history

✅ **Answer: B**

**Explanation:** PATH decides where shell finds executable commands.

---





## **Process Management — IMP MCQs (Exam Ready)**

### **Basics: Process / Program / Thread**

#### **Q1. A process is:**

- A) A program stored on disk
- B) A program in execution with state and resources
- C) A CPU instruction
- D) A file system directory

 **Answer: B**

**Explanation:** Process = active program with execution context.

---

#### **Q2. Which is NOT typically part of a process?**

- A) Heap
- B) Stack
- C) Registers
- D) Compiler

 **Answer: D**

**Explanation:** Compiler is a tool, not part of process memory/context.

---

#### **Q3. Program vs Process: correct statement is:**

- A) Program is active, process is passive
- B) Process is passive, program is active
- C) Program is passive on disk, process is active in memory
- D) Both are same always

 **Answer: C**

**Explanation:** Program = static; Process = executing instance.

---

#### **Q4. Process vs Thread: correct statement is:**

- A) Thread has separate memory space always
- B) Threads share memory within the same process
- C) Process context switch is faster than thread
- D) Thread cannot improve responsiveness

 **Answer: B**

**Explanation:** Threads share process address space.

---

#### **Q5. Compared to processes, thread context switching is:**

- A) Slower
- B) Faster

- C) Same speed always
- D) Impossible

✓ **Answer: B**

**Explanation:** Thread is lightweight → less overhead.

---

### Process States

**Q6. Correct process state sequence is:**

- A) New → Running → Ready → Blocked → Terminated
- B) New → Ready → Running → Blocked → Terminated
- C) Ready → New → Running → Terminated → Blocked
- D) New → Blocked → Ready → Running → Terminated

✓ **Answer: B**

**Explanation:** Standard flow: New → Ready → Running → Blocked → Ready → ... → Terminated.

---

**Q7. A process waiting for I/O is in:**

- A) Ready
- B) Running
- C) Blocked
- D) Terminated

✓ **Answer: C**

**Explanation:** Blocked = waiting for event/resource (I/O).

---

**Q8. Which states occur only once in process lifetime?**

- A) Ready and Running
- B) Blocked and Running
- C) New and Terminated
- D) Ready and Blocked

✓ **Answer: C**

**Explanation:** Creation and exit happen once.

---

### Schedulers (Very High Weight)

**Q9. Long-term scheduler mainly controls:**

- A) CPU allocation
- B) Degree of multiprogramming
- C) Deadlock recovery
- D) Disk scheduling

✓ **Answer: B**

**Explanation:** Decides how many processes enter memory (ready queue).

---

**Q10. Which scheduler runs most frequently?**

- A) Long-term
- B) Medium-term
- C) Short-term
- D) None

✓ **Answer: C**

**Explanation:** Short-term (CPU scheduler) runs very frequently.

---

**Q11. Medium-term scheduler is also called:**

- A) CPU scheduler
- B) Swapper
- C) Job scheduler
- D) Dispatcher

✓ **Answer: B**

**Explanation:** It swaps processes between RAM and disk.

---

**Q12. Short-term scheduler selects process from:**

- A) Job pool
- B) Suspended queue
- C) Ready queue
- D) Disk partition table

✓ **Answer: C**

**Explanation:** Picks from ready queue and gives CPU.

---

**Preemptive vs Non-preemptive**

**Q13. In non-preemptive scheduling, CPU is released when:**

- A) Another process arrives
- B) OS forcibly interrupts
- C) Process finishes or blocks
- D) Time quantum ends

✓ **Answer: C**

**Explanation:** No forced removal from CPU.

---

**Q14. Which scheduling is preemptive by nature?**

- A) FCFS
- B) Round Robin
- C) Non-preemptive SJF
- D) Batch scheduling

✓ **Answer: B**

**Explanation:** RR uses time quantum and preemption.

---

**Q15. SRTF stands for:**

- A) Shortest Ready Task First
- B) Shortest Remaining Time First
- C) System Round Time First
- D) Scheduled Response Time Factor

✓ **Answer: B**

**Explanation:** Preemptive version of SJF.

---

**CPU Scheduling Algorithms (Core MCQs)**

**Q16. FCFS scheduling follows:**

- A) LIFO
- B) FIFO
- C) Priority queue
- D) Random order

✓ **Answer: B**

**Explanation:** FCFS = First-In First-Out.

---

**Q17. Major drawback of FCFS is:**

- A) Starvation
- B) Convoy effect
- C) Deadlock always
- D) Thrashing always

✓ **Answer: B**

**Explanation:** Long job delays all short jobs behind it.

---

**Q18. Which algorithm gives minimum average waiting time (theoretically)?**

- A) FCFS
- B) SJF
- C) Round Robin
- D) Priority scheduling

✓ **Answer: B**

**Explanation:** SJF minimizes average waiting time (if burst times known).

---

**Q19. Main disadvantage of SJF is:**

- A) Too many context switches
- B) Requires prior burst time knowledge + starvation for long jobs
- C) Works only for I/O bound jobs
- D) Always increases waiting time

✓ **Answer: B**

**Explanation:** Long jobs may starve; burst time estimation is tough.

---

**Q20. Priority scheduling mainly suffers from:**

- A) Convoy effect
- B) Starvation
- C) No CPU utilization
- D) Always non-preemptive

✓ **Answer: B**

**Explanation:** Low priority may wait indefinitely.

---

**Q21. Starvation in priority scheduling is solved using:**

- A) Swapping
- B) Aging
- C) Spooling
- D) Paging

✓ **Answer: B**

**Explanation:** Aging increases priority gradually over time.

---

**Q22. In Round Robin, time slice is called:**

- A) CPU burst
- B) Time Quantum
- C) Turnaround time
- D) Response delay

✓ **Answer: B**

**Explanation:** Each process gets fixed quantum.

---

**Q23. If time quantum is too small in RR:**

- A) It behaves like FCFS
- B) Too many context switches occur
- C) Starvation becomes impossible
- D) CPU becomes idle always

✅ **Answer: B**

**Explanation:** Small quantum increases overhead.

---

**Q24. If time quantum is too large in RR:**

- A) Overhead becomes infinite
- B) It behaves like FCFS
- C) It becomes SJF
- D) It becomes Priority scheduling

✅ **Answer: B**

**Explanation:** Large quantum → near non-preemptive FCFS behavior.

---

**Q25. In Multilevel Queue scheduling:**

- A) Only one queue exists
- B) Each queue can have its own scheduling algorithm
- C) Scheduling is always RR
- D) It guarantees minimum waiting time always

✅ **Answer: B**

**Explanation:** Different categories → different policies.

---

**Performance Metrics (Must-Memorize Formulas)**

**Q26. Turnaround Time is:**

- A) Arrival – Burst
- B) Completion – Arrival
- C) Burst – Arrival
- D) Completion – Burst

✅ **Answer: B**

**Explanation:**  $TAT = CT - AT$ .

---

**Q27. Waiting Time can be calculated as:**

- A) Burst – Turnaround
- B) Arrival – Completion
- C) Turnaround – Burst
- D) Completion – Burst

✓ **Answer: C**

**Explanation:**  $WT = TAT - BT$ .

---

**Q28. Response Time means:**

- A) Total execution time on CPU
- B) Time from arrival until first CPU allocation
- C) Time spent in I/O
- D) Completion time only

✓ **Answer: B**

**Explanation:** Response = first start time – arrival time.

---

**Belady's Anomaly (Syllabus Required)**

**Q29. Belady's anomaly occurs in:**

- A) LRU Page Replacement
- B) Optimal Page Replacement
- C) FIFO Page Replacement
- D) SJF Scheduling

✓ **Answer: C**

**Explanation:** Only FIFO can increase page faults when frames increase.

---

**Q30. Belady's anomaly does NOT occur in:**

- A) FIFO
- B) LRU
- C) Both FIFO and LRU
- D) Only FIFO and Optimal

✓ **Answer: B**

**Explanation:** LRU is a stack algorithm → no anomaly.

---

**Process Creation System Calls (fork/exec/waitpid) — Most Important**

**Q31. fork() is used to:**

- A) Replace a process with a new program
- B) Create a duplicate child process
- C) Kill a running process
- D) Suspend a process

✓ **Answer: B**

**Explanation:** fork() creates child = copy of parent.

---

**Q32. fork() return value in child process is:**

- A) -1
- B) 0
- C) Child PID
- D) Parent PID

☒ **Answer: B**

**Explanation:** 0 in child, PID in parent.

---

**Q33. exec() does:**

- A) Creates a new child process
- B) Replaces current process memory with a new program
- C) Waits for child
- D) Terminates parent

☒ **Answer: B**

**Explanation:** exec loads new program into same process (same PID).

---

**Q34. After exec(), which remains same?**

- A) Process ID (PID)
- B) Program code
- C) Data segment
- D) Heap content

☒ **Answer: A**

**Explanation:** PID remains same; memory is replaced.

---

**Q35. waitpid() is used for:**

- A) Parent waits for specific child + collects exit status
- B) Child waits for parent
- C) Scheduling CPU jobs
- D) Creating threads

☒ **Answer: A**

**Explanation:** Prevents zombies and synchronizes execution.

---

**Orphan vs Zombie (Very Common Trap)**

**Q36. Orphan process occurs when:**

- A) Child terminates before parent
- B) Parent terminates while child still running



- C) Parent waits for child
- D) Child is suspended

✓ **Answer: B**

**Explanation:** Child becomes orphan and is adopted by init/systemd.

---

**Q37. Zombie process occurs when:**

- A) Parent dies first
- B) Child never terminates
- C) Child terminates but parent does not collect exit status
- D) Process is swapped out

✓ **Answer: C**

**Explanation:** Zombie = “defunct” entry remains in process table.

---

**Q38. Zombie process exists mainly to store:**

- A) CPU burst time
- B) Child exit status
- C) Program code
- D) Kernel stack

✓ **Answer: B**

**Explanation:** Wait/Waitpid collects it later.

---

**Threads + IPC**

**Q39. Threads in the same process share:**

- A) Separate stack and separate memory
- B) Same memory, same code, same data
- C) Different PID always
- D) Separate file systems

✓ **Answer: B**

**Explanation:** Shared address space; separate stack/registers.

---

**Q40. IPC is required mainly because:**

- A) Processes are isolated with separate memory spaces
- B) Threads cannot communicate
- C) Kernel forbids communication
- D) CPU cannot run multiple processes

✓ **Answer: A**

**Explanation:** Process isolation requires IPC mechanisms.

## ✓ Memory Management — IMP MCQs (CCEE Focused)

### Memory Hierarchy + Properties

**Q1. Correct memory hierarchy from fastest to slowest:**

- A) HDD → RAM → Cache → Register
- B) Register → Cache → RAM → SSD/HDD
- C) Cache → Register → RAM → HDD
- D) RAM → Register → Cache → SSD

✓ **Answer: B**

**Explanation:** Registers are fastest; disk is slowest.

---

**Q2. As we move upward in memory hierarchy:**

- A) Space increases, speed decreases
- B) Space decreases, speed increases
- C) Cost decreases, speed decreases
- D) Space increases, cost increases

✓ **Answer: B**

**Explanation:** Upwards → smaller but faster memory.

---

**Q3. Which memory is fastest but smallest and costliest?**

- A) RAM
- B) SSD
- C) Cache
- D) Registers

✓ **Answer: D**

**Explanation:** Registers have highest speed and cost/bit.

---

### Need for Memory Management

**Q4. Main purpose of memory management is to:**

- A) Remove viruses
- B) Allocate, track and protect memory for processes
- C) Improve screen resolution
- D) Create new users

✓ **Answer: B**

**Explanation:** OS must allocate RAM, track free/used and prevent overwriting.

---

**Q5. Without memory management, one major risk is:**

- A) CPU stops permanently
- B) Processes may overwrite each other's memory
- C) Hard disk becomes faster
- D) Linux cannot boot

✓ **Answer: B**

**Explanation:** No protection → unstable system.

---

### **Key Concepts: Swapping / Cache / Purging / Archival**

#### **Q6. Swapping means:**

- A) Sorting memory blocks
- B) Moving data between RAM and disk
- C) Encrypting files
- D) Cleaning cache

✓ **Answer: B**

**Explanation:** Swapping transfers processes/pages between RAM and disk.

---

#### **Q7. Cache memory is best described as:**

- A) Permanent storage
- B) High-speed buffer storing frequently accessed data
- C) A file system directory
- D) A scheduler

✓ **Answer: B**

**Explanation:** Cache reduces memory access time.

---

#### **Q8. Archival means:**

- A) Moving data upward only
- B) Moving data downward only for permanent storage
- C) Moving data between CPU and registers
- D) Deleting data automatically

✓ **Answer: B**

**Explanation:** Archival = downward movement for long-term storage.

---

### **Locality of Reference (Very High Weight)**

#### **Q9. Locality of reference means:**

- A) CPU accesses memory randomly always
- B) Programs tend to reuse same/nearby memory locations often

- C) Cache is always empty
- D) Paging is not required

✓ **Answer: B**

**Explanation:** Programs show predictable access patterns.

---

**Q10. Temporal locality means:**

- A) Nearby memory locations are accessed soon
- B) Same memory location is accessed again soon
- C) Only code locality exists
- D) Disk access becomes faster

✓ **Answer: B**

**Explanation:** Reuse of the same data/instruction (loops, functions).

---

**Q11. Spatial locality means:**

- A) Same location accessed repeatedly
- B) Nearby memory locations accessed soon
- C) Only heap is accessed
- D) No locality exists

✓ **Answer: B**

**Explanation:** Arrays and sequential instructions create spatial locality.

---

**Fragmentation (Very Common Exam Trap)**

**Q12. Internal fragmentation is:**

- A) Free space scattered outside allocated blocks
- B) Wasted space inside allocated memory block
- C) Only occurs in disks
- D) Same as compaction

✓ **Answer: B**

**Explanation:** Fixed allocation causes unused space inside a block.

---

**Q13. External fragmentation is:**

- A) Memory wastage inside a block
- B) Free memory scattered into small non-contiguous holes
- C) Always removed by paging
- D) Caused by CPU scheduling

✓ **Answer: B**

**Explanation:** Total free memory exists but not as one contiguous block.

---

**Q14. External fragmentation is mostly seen in:**

- A) Paging
- B) Variable partition allocation
- C) Registers only
- D) TLB caching

☒ **Answer: B**

**Explanation:** Variable-sized contiguous allocation causes holes.

---

**Q15. Paging mainly suffers from:**

- A) External fragmentation
- B) Internal fragmentation
- C) No fragmentation
- D) Only disk fragmentation

☒ **Answer: B**

**Explanation:** Last page may not fully fill → internal waste.

---

**Dynamic Storage Allocation Algorithms**

**Q16. First Fit strategy allocates:**

- A) Smallest block that fits
- B) Largest block that fits
- C) First available block that fits
- D) Random block

☒ **Answer: C**

**Explanation:** First suitable block is selected.

---

**Q17. Best Fit strategy allocates:**

- A) First suitable block
- B) Smallest suitable block
- C) Largest suitable block
- D) Middle block

☒ **Answer: B**

**Explanation:** Best fit reduces leftover space but may create tiny holes.

---

**Q18. Worst Fit strategy allocates:**

- A) Largest available block
- B) Smallest available block

- C) First available block
- D) Exact-size block only

✓ **Answer: A**

**Explanation:** Uses largest hole to leave big remaining holes.

---

### **Compaction**

**Q19. Compaction is used to reduce:**

- A) Internal fragmentation
- B) External fragmentation
- C) Paging overhead
- D) TLB misses

✓ **Answer: B**

**Explanation:** Compaction merges scattered holes into one big free block.

---

**Q20. Compaction requires:**

- A) No relocation support
- B) Relocation capability (address adjustment)
- C) TLB only
- D) FIFO scheduling

✓ **Answer: B**

**Explanation:** Processes must be movable in memory.

---

### **Paging (Most Important Section)**

**Q21. In paging:**

- A) Logical memory is divided into frames
- B) Physical memory is divided into pages
- C) Logical memory is divided into pages and physical memory into frames
- D) Both are divided into segments

✓ **Answer: C**

**Explanation:** Pages (logical) map to frames (physical).

---

**Q22. Page size and frame size are:**

- A) Always different
- B) Always equal
- C) Frame is always bigger
- D) Page is always bigger

✓ **Answer: B**

**Explanation:** Equal size makes mapping easy.

---

**Q23. The data structure that maps page number to frame number is:**

- A) Segment table
- B) Page table
- C) Ready queue
- D) FAT

✓ **Answer: B**

**Explanation:** Page table does logical → physical mapping.

---

**Q24. Address translation in paging is performed by:**

- A) CPU scheduler
- B) MMU (Memory Management Unit)
- C) Hard disk controller
- D) Shell

✓ **Answer: B**

**Explanation:** MMU translates logical to physical addresses.

---

**Q25. TLB is used to:**

- A) Store complete processes
- B) Speed up address translation by caching page table entries
- C) Replace page table completely
- D) Remove internal fragmentation

✓ **Answer: B**

**Explanation:** TLB hit is fast; miss requires memory access.

---

**Q26. Dirty bit in page table indicates:**

- A) Page is empty
- B) Page has been modified
- C) Page is read-only
- D) Page is in TLB

✓ **Answer: B**

**Explanation:** Dirty=1 → must write back to disk when replacing.

---

**Q27. One major advantage of paging is:**

- A) Eliminates external fragmentation
- B) Eliminates internal fragmentation completely
- C) Requires no hardware support
- D) Uses only contiguous memory

✓ **Answer: A**

**Explanation:** Paging removes external fragmentation.

---

### Shared Pages + Re-entrant Code

**Q28. Shared pages are commonly used for:**

- A) Writable heap memory
- B) Read-only libraries/code shared across processes
- C) Private stack memory
- D) Process table entries

✓ **Answer: B**

**Explanation:** Shared libraries reduce memory usage.

---

**Q29. Re-entrant code is:**

- A) Self-modifying code
- B) Read-only code executable by multiple processes concurrently
- C) Code that runs only once
- D) Code that always writes into memory

✓ **Answer: B**

**Explanation:** Pure/re-entrant code can be shared safely.

---

### Throttling + Thrashing

**Q30. Throttling is used when:**

- A) Disk is full
- B) Page-fault rate becomes too high (thrashing risk)
- C) CPU is overheating
- D) User enters wrong password

✓ **Answer: B**

**Explanation:** OS reduces active processes to stabilize memory usage.

---

### Segmentation (Important vs Paging Difference)

**Q31. Segmentation divides memory into:**

- A) Fixed-size pages
- B) Fixed-size frames



- C) Logical variable-size segments (code/data/stack)
- D) Only equal-size blocks

✓ **Answer: C**

**Explanation:** Segments are logical units and variable-sized.

---

**Q32. Segment table entry contains:**

- A) Page number and frame number
- B) Base address and limit
- C) Only page offset
- D) Only PID

✓ **Answer: B**

**Explanation:** Base + limit define the segment region.

---

**Q33. A logical address in segmentation is:**

- A) (Frame number, offset)
- B) (Page number, frame number)
- C) (Segment number, offset)
- D) (Base, limit)

✓ **Answer: C**

**Explanation:** CPU uses segment number and offset.

---

**Q34. In segmentation, address is valid if:**

- A) Offset > limit
- B) Offset < limit
- C) Segment number is zero
- D) Base address is zero

✓ **Answer: B**

**Explanation:** If offset exceeds limit → memory fault.

---

**Virtual Memory (Intro Level)**

**Q35. Virtual memory allows:**

- A) Programs larger than RAM to execute using disk
- B) CPU to run without OS
- C) Only real-time processes
- D) No need of paging

✓ **Answer: A**

**Explanation:** Disk extends RAM conceptually via paging.

---

**Q36. Virtual memory mainly uses:**

- A) Demand paging and page replacement
- B) Only compaction
- C) Only segmentation
- D) Only swapping entire processes always

☒ **Answer: A**

**Explanation:** Loads pages on demand; replaces when needed.

---



## ✔ Virtual Memory & Page Replacement — IMP MCQs (CCEE Ready)

### Virtual Memory + Demand Paging

#### Q1. Virtual memory allows:

- A) Programs to run only if fully loaded in RAM
- B) Logical memory to be larger than physical memory
- C) No need of disk storage
- D) CPU to execute without OS

✔ **Answer: B**

**Explanation:** Virtual memory gives illusion of large memory using disk + RAM.

---

#### Q2. Virtual memory mainly works using:

- A) Demand paging
- B) File compression
- C) CPU scheduling
- D) Disk formatting

✔ **Answer: A**

**Explanation:** Demand paging loads pages only when required.

---

#### Q3. Demand Paging means:

- A) Load entire program into RAM before execution
- B) Load pages into RAM only when needed
- C) Always keep all pages in cache
- D) Load only stack segment

✔ **Answer: B**

**Explanation:** Pages are loaded on demand → saves memory and time.

---

#### Q4. A major benefit of Demand Paging is:

- A) Increases memory usage
- B) Increases load time
- C) Reduces wasted memory and improves utilization
- D) Removes page faults permanently

✔ **Answer: C**

**Explanation:** Only required pages are loaded → less RAM usage.

---

### Page Fault (Very Important Steps)

#### Q5. A page fault occurs when:

- A) CPU finds page in RAM
- B) Requested page is not present in physical memory
- C) Page is found in TLB
- D) RAM is empty

✅ **Answer: B**

**Explanation:** Page fault happens when page is not currently in RAM.

---

**Q6. Correct order during page fault handling is:**

- A) OS updates page table → CPU checks MMU → page loaded
- B) MMU checks page table → page not present → OS loads from disk
- C) CPU ignores fault → process continues
- D) Page is deleted permanently

✅ **Answer: B**

**Explanation:** MMU detects missing page → OS loads it from disk.

---

**Q7. If RAM is full during page fault, OS must:**

- A) Stop the CPU permanently
- B) Run a page replacement algorithm
- C) Delete the process
- D) Increase cache size

✅ **Answer: B**

**Explanation:** Page replacement selects victim page to evict.

---

**Valid / Invalid Bit**

**Q8. Valid bit = 1 means:**

- A) Page not in RAM
- B) Page exists in RAM
- C) Illegal page
- D) Page is corrupted

✅ **Answer: B**

**Explanation:** Valid=1 → present in memory.

---

**Q9. Invalid bit = 0 can mean:**

- A) Page is present in RAM
- B) Page not present OR illegal access
- C) Page is always dirty
- D) Page must be replaced

✓ **Answer: B**

**Explanation:** Invalid indicates not present or invalid reference.

---

### **Effective Access Time (Concept)**

**Q10. Effective memory access time increases when:**

- A) Page fault probability decreases
- B) Page fault probability increases
- C) CPU speed increases
- D) Cache is disabled

✓ **Answer: B**

**Explanation:** Page fault handling time is huge → EAT rises quickly.

---

### **Page Replacement Algorithms (Top Exam Area)**

**Q11. FIFO replacement removes:**

- A) Least recently used page
- B) Oldest page in memory
- C) Page used least frequently
- D) Page with highest priority

✓ **Answer: B**

**Explanation:** FIFO evicts earliest loaded page.

---

**Q12. FIFO suffers from:**

- A) Thrashing always
- B) Belady's anomaly
- C) No page faults
- D) Internal fragmentation

✓ **Answer: B**

**Explanation:** FIFO can increase faults even with more frames.

---

**Q13. Belady's anomaly means:**

- A) More frames always reduce page faults
- B) More frames may increase page faults in FIFO
- C) LRU gives maximum faults
- D) Optimal is worst algorithm

✓ **Answer: B**

**Explanation:** Happens only in FIFO.

---

**Q14. Optimal page replacement removes the page:**

- A) Used most recently
- B) Not used for the longest time in the future
- C) Least frequently used
- D) Oldest loaded page

☒ **Answer: B**

**Explanation:** Uses future knowledge (best theoretical).

---

**Q15. Optimal replacement is not practical because:**

- A) It causes Belady's anomaly
- B) It requires future knowledge of references
- C) It has too many faults
- D) It is always slower than FIFO

☒ **Answer: B**

**Explanation:** Real OS cannot predict future page access.

---

**Q16. LRU replacement removes the page:**

- A) Least recently used in past
- B) First loaded page
- C) Most frequently used page
- D) Not used in future

☒ **Answer: A**

**Explanation:** LRU uses past history and locality principle.

---

**Q17. LRU is based on:**

- A) Deadlock detection
- B) Locality of reference
- C) Round Robin scheduling
- D) Convoy effect

☒ **Answer: B**

**Explanation:** Recently used pages likely used again soon.

---

**Q18. Which algorithm does NOT suffer from Belady's anomaly?**

- A) FIFO
- B) LRU
- C) FIFO and LRU both
- D) Only FIFO

✓ **Answer: B**

**Explanation:** LRU is a stack algorithm, so no anomaly.

---

**Q19. Clock / Second Chance algorithm is:**

- A) Same as Optimal
- B) FIFO improved using reference bit
- C) Same as MRU
- D) Same as LFU

✓ **Answer: B**

**Explanation:** Gives pages a second chance before eviction.

---

**Q20. LFU removes the page with:**

- A) Least recently used time
- B) Least frequency count
- C) Oldest load time
- D) Highest frequency count

✓ **Answer: B**

**Explanation:** LFU uses frequency of access.

---

**Q21. MRU removes the page:**

- A) Most recently used
- B) Least frequently used
- C) Oldest in queue
- D) Least recently used

✓ **Answer: A**

**Explanation:** Opposite of LRU (useful in some special workloads).

---

**Dirty Bit (Must Know)**

**Q22. Dirty bit = 1 means:**

- A) Page is read-only
- B) Page was modified in RAM
- C) Page is not present
- D) Page is invalid

✓ **Answer: B**

**Explanation:** Modified page must be written back before eviction.

---

**Q23. If a page is clean (dirty bit = 0), then on replacement:**

- A) It must always be written to disk
- B) It can be discarded without writing back
- C) It becomes invalid permanently
- D) It becomes a TLB hit

✓ **Answer: B**

**Explanation:** No need to save because no changes.

---

### **TLB (Very Common MCQ)**

#### **Q24. TLB is:**

- A) A disk scheduler
- B) A fast cache storing recent page table entries
- C) A process scheduler
- D) A file system table

✓ **Answer: B**

**Explanation:** Speeds up address translation.

---

#### **Q25. TLB Hit means:**

- A) Page not found anywhere
- B) Translation found in TLB (fast access)
- C) Page must be replaced
- D) RAM is full

✓ **Answer: B**

**Explanation:** Hit avoids page table memory access.

---

#### **Q26. TLB Miss means:**

- A) CPU crashes
- B) Page is not in RAM always
- C) Page table lookup in memory is required
- D) Page is invalid always

✓ **Answer: C**

**Explanation:** Miss → access page table (slower than hit).

---

### **Page Sharing + Re-entrant Code**

#### **Q27. Shared pages are typically:**

- A) Writable user stack shared by processes
- B) Read-only library/code shared among processes



- C) Different for every process always
- D) Only for kernel memory

✓ **Answer: B**

**Explanation:** Shared libraries save memory.

---

**Q28. Re-entrant code is:**

- A) Code that modifies itself
- B) Code that can be shared safely (read-only)
- C) Code that runs only once
- D) Code that always causes page faults

✓ **Answer: B**

**Explanation:** Safe for concurrent execution by multiple processes.

---

**Thrashing + Throttling + Working Set**

**Q29. Thrashing happens when:**

- A) CPU spends more time executing than paging
- B) Page faults are so frequent that swapping dominates execution
- C) Disk becomes faster
- D) TLB hit ratio becomes 100%

✓ **Answer: B**

**Explanation:** CPU wastefully handles page faults and I/O.

---

**Q30. Thrashing causes:**

- A) Increased CPU utilization
- B) Decreased CPU utilization
- C) No change in performance
- D) Faster execution

✓ **Answer: B**

**Explanation:** CPU waits while paging happens.

---

**Q31. Throttling is used to:**

- A) Increase number of processes
- B) Reduce number of running processes to control thrashing
- C) Remove page tables
- D) Increase page fault rate

✓ **Answer: B**

**Explanation:** OS suspends some processes to reduce faults.

---

**Q32. Working set refers to:**

- A) All pages of all processes
- B) Set of actively used pages of a process
- C) Only pages in disk
- D) Only code pages

✅ **Answer: B**

**Explanation:** If working set fits in RAM → smooth execution.

---

**Demand Paging vs Pre-Paging**

**Q33. Demand paging loads:**

- A) Multiple pages in advance
- B) Pages only when required
- C) All pages at startup always
- D) No pages ever

✅ **Answer: B**

**Explanation:** Demand = “load when needed”.

---

**Q34. Pre-paging loads:**

- A) Only the demanded page
- B) Multiple pages at once to reduce future faults
- C) Only stack segment
- D) Only dirty pages

✅ **Answer: B**

**Explanation:** Pre-paging reduces page fault frequency but may waste memory.

---

**Advantages / Disadvantages**

**Q35. Major advantage of virtual memory is:**

- A) Programs must fit in RAM always
- B) Logical memory can exceed physical RAM
- C) Page faults become impossible
- D) CPU scheduling is removed

✅ **Answer: B**

**Explanation:** Allows large programs and multiprogramming.

---

**Q36. One disadvantage of virtual memory is:**

- A) No multitasking
- B) More page faults can reduce performance
- C) No MMU required
- D) No need of disk

✅ **Answer: B**

**Explanation:** Excess faults → slowdown and thrashing.

---



## **Deadlocks, Synchronization & Process Coordination — IMP MCQs**

### **Deadlock Basics**

**Q1. Deadlock is a condition where:**

- A) CPU becomes idle
- B) Two or more processes wait forever for resources held by each other
- C) A process finishes quickly
- D) Only one process runs

 **Answer: B**

**Explanation:** Deadlock = circular waiting → no process can proceed.

---

**Q2. Deadlock occurs only if:**

- A) At least one process is running
- B) All Coffman conditions hold simultaneously
- C) CPU utilization is high
- D) Memory is full

 **Answer: B**

**Explanation:** All 4 necessary conditions must hold.

---

### **Coffman Conditions (Very High Weight)**

**Q3. Mutual exclusion means:**

- A) Resources can be shared by all processes
- B) Only one process can use a resource at a time
- C) A process releases resources immediately
- D) CPU is shared equally

 **Answer: B**

**Explanation:** Non-shareable resource → one process at a time.

---

**Q4. Hold and wait means:**

- A) A process holds resources while waiting for additional resources
- B) A process holds no resource
- C) A process releases resources before waiting
- D) CPU waits for memory

 **Answer: A**

**Explanation:** Process keeps acquired resources while requesting more.

---

**Q5. No pre-emption means:**

- A) OS can forcibly take resources anytime
- B) CPU can be preempted by scheduler
- C) Resources cannot be forcibly taken; they must be released voluntarily
- D) Paging is not possible

✓ **Answer: C**

**Explanation:** Resource is released only by the process itself.

---

**Q6. Circular wait means:**

- A) Processes never request resources
- B) Processes form a cycle of waiting for each other's resources
- C) CPU executes in circular queue
- D) A process waits only once

✓ **Answer: B**

**Explanation:** P1 waits for P2, P2 waits for P3 ... Pn waits for P1.

---

**Q7. Deadlock can be prevented by:**

- A) Ensuring all 4 Coffman conditions always occur
- B) Violating at least one Coffman condition
- C) Increasing CPU speed
- D) Increasing disk space

✓ **Answer: B**

**Explanation:** If any one condition is broken → no deadlock.

---

**Deadlock Handling Approaches**

**Q8. Which is NOT a deadlock handling method?**

- A) Prevention
- B) Avoidance
- C) Detection & Recovery
- D) Scheduling

✓ **Answer: D**

**Explanation:** Scheduling is not a deadlock strategy.

---

**Q9. Deadlock prevention works by:**

- A) Allowing deadlock and then fixing it later
- B) Ensuring system always remains in safe state
- C) Preventing one of the necessary deadlock conditions
- D) Running Banker's algorithm

✓ **Answer: C**

**Explanation:** Prevention breaks mutual exclusion / hold-wait / circular wait / no-preemption.

---

**Q10. Deadlock avoidance means:**

- A) Deadlock is ignored
- B) Deadlock is prevented by breaking all conditions
- C) Resources are allocated only if system remains safe
- D) Deadlocks are solved by killing processes immediately

✓ **Answer: C**

**Explanation:** Avoidance checks safe state before granting request.

---

**Q11. The key concept behind avoidance is:**

- A) Starvation
- B) Safe state
- C) Thrashing
- D) Paging

✓ **Answer: B**

**Explanation:** Safe state guarantees a safe sequence exists.

---

**Q12. Banker's Algorithm is used for:**

- A) Deadlock prevention
- B) Deadlock avoidance
- C) Deadlock detection
- D) Thread synchronization only

✓ **Answer: B**

**Explanation:** Banker's ensures safe state before allocation.

---

**Q13. Banker's algorithm requires:**

- A) No knowledge of maximum resource needs
- B) Maximum resource demand known in advance
- C) Only single process exists
- D) Paging must be disabled

✓ **Answer: B**

**Explanation:** Maximum resource needs must be known → limits practicality.

---

**Q14. Deadlock detection & recovery means:**

- A) Never allowing deadlock
- B) Allow deadlock, detect cycle, then recover
- C) Using only prevention techniques
- D) Using only FIFO scheduling

✅ **Answer: B**

**Explanation:** Deadlock allowed → detection + recovery strategy used.

---

**Q15. A cycle in Resource Allocation Graph indicates:**

- A) Deadlock always in all cases
- B) Deadlock in systems with single instance per resource
- C) No deadlock possible
- D) Only starvation

✅ **Answer: B**

**Explanation:** With single instance of each resource → cycle = deadlock.

---

**Deadlock vs Starvation (Common Confusion MCQ)**

**Q16. Deadlock means:**

- A) One process waits forever, others proceed normally
- B) All involved processes are blocked permanently due to cyclic waiting
- C) CPU scheduling is unfair
- D) Memory becomes full

✅ **Answer: B**

**Explanation:** Deadlock = complete stop for involved processes.

---

**Q17. Starvation means:**

- A) Circular dependency exists
- B) One process waits indefinitely because others get resources repeatedly
- C) All processes stop
- D) Deadlock always occurs

✅ **Answer: B**

**Explanation:** System continues but one process never gets chance.

---

**Q18. A common solution to starvation is:**

- A) FIFO
- B) Aging
- C) Paging
- D) Thrashing

☒ **Answer: B**

**Explanation:** Aging increases priority gradually.

---

### **Process Synchronization + Race Condition**

**Q19. Process synchronization is required when:**

- A) Only one process exists
- B) Multiple processes/threads share data/resources
- C) RAM is full
- D) CPU is idle

☒ **Answer: B**

**Explanation:** Concurrent access can corrupt data.

---

**Q20. A race condition occurs when:**

- A) Program output depends on execution timing/order of threads
- B) Only one thread executes
- C) OS crashes
- D) Scheduling is FCFS

☒ **Answer: A**

**Explanation:** Unpredictable interleaving → inconsistent results.

---

### **Semaphore (Very High Weight)**

**Q21. Semaphore was introduced by:**

- A) Linus Torvalds
- B) Dijkstra
- C) Dennis Ritchie
- D) Charles Babbage

☒ **Answer: B**

**Explanation:** Semaphores were proposed by Dijkstra.

---

**Q22. Semaphore is mainly used to:**

- A) Speed up compilation
- B) Control access to shared resources
- C) Reduce disk space
- D) Create processes

☒ **Answer: B**

**Explanation:** Synchronization tool for critical sections.

---



**Q23. wait() / P() operation does:**

- A) Increments semaphore value
- B) Decrements semaphore value (or blocks if not possible)
- C) Deletes semaphore
- D) Terminates process

☒ **Answer: B**

**Explanation:** wait() decreases S; if  $S \leq 0 \rightarrow$  process waits.

---

**Q24. signal() / V() operation does:**

- A) Decrements semaphore value
- B) Increments semaphore value
- C) Creates new process
- D) Causes deadlock always

☒ **Answer: B**

**Explanation:** signal() increases S and may wake waiting process.

---

**Q25. Semaphore operations must be:**

- A) Non-atomic
- B) Atomic
- C) Executed only in user mode
- D) Executed only after scheduling

☒ **Answer: B**

**Explanation:** If not atomic  $\rightarrow$  synchronization fails.

---

**Q26. A binary semaphore can take values:**

- A) 0 and 1 only
- B) Any integer
- C) Only negative values
- D) Only prime numbers

☒ **Answer: A**

**Explanation:** Binary semaphore is like lock/unlock.

---

**Q27. A counting semaphore is used when:**

- A) Only one process should enter
- B) Multiple instances of a resource exist
- C) CPU scheduling is required
- D) Paging is enabled

✓ **Answer: B**

**Explanation:** Example: 10 printers → semaphore initialized to 10.

---

### **Mutex vs Semaphore**

**Q28. Mutex is:**

- A) A type of file system
- B) A strict binary lock for mutual exclusion
- C) A CPU scheduling algorithm
- D) A page replacement technique

✓ **Answer: B**

**Explanation:** Mutex ensures only one thread enters critical section.

---

**Q29. Key difference: Mutex vs Semaphore**

- A) Mutex can be counting
- B) Semaphore has ownership rules
- C) Only owner can unlock mutex, semaphore can be signaled by any process
- D) Both are identical always

✓ **Answer: C**

**Explanation:** Mutex enforces ownership; semaphore is general synchronization.

---

### **Critical Section Problem (3 Conditions)**

**Q30. Mutual exclusion in critical section means:**

- A) Many processes enter simultaneously
- B) Only one process enters critical section at a time
- C) No process enters ever
- D) CPU becomes idle

✓ **Answer: B**

**Explanation:** Prevents simultaneous update of shared data.

---

**Q31. Progress condition ensures:**

- A) Only OS makes decisions always
- B) If no one is in CS, selection of next process cannot be postponed forever
- C) CPU never switches
- D) Starvation always occurs

✓ **Answer: B**

**Explanation:** Decision must be made in finite time.

---

**Q32. Bounded waiting means:**

- A) A process may wait forever
- B) A limit exists on how long a process waits before entering CS
- C) CPU is always busy
- D) Deadlock always happens

☒ **Answer: B**

**Explanation:** Prevents starvation in critical section entry.

---

**Producer–Consumer (Bounded Buffer) — Most Asked Problem**

**Q33. Producer-consumer problem is also called:**

- A) Dining philosophers
- B) Bounded buffer problem
- C) Readers-writers
- D) Thrashing problem

☒ **Answer: B**

**Explanation:** Fixed-size buffer shared by producer and consumer.

---

**Q34. In producer-consumer, empty semaphore represents:**

- A) Filled slots in buffer
- B) Empty slots in buffer
- C) Mutex lock
- D) CPU time quantum

☒ **Answer: B**

**Explanation:** empty = number of available empty slots.

---

**Q35. In producer-consumer, full semaphore represents:**

- A) Number of filled slots
- B) Number of empty slots
- C) Process priority
- D) Page faults

☒ **Answer: A**

**Explanation:** full = available filled items for consumer.

---

**Q36. mutex semaphore is used for:**

- A) Counting buffer slots
- B) Ensuring mutual exclusion on buffer access

- C) Page replacement
- D) Deadlock detection

✅ **Answer: B**

**Explanation:** mutex prevents simultaneous access to shared buffer.

---

### Dining Philosophers (Conceptual MCQ)

**Q37. Dining philosophers problem mainly demonstrates:**

- A) Paging
- B) Deadlock and starvation in synchronization
- C) CPU scheduling only
- D) File allocation

✅ **Answer: B**

**Explanation:** Incorrect resource acquisition leads to deadlock/starvation.

---

